Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 3: Laplace transforms

How Topic 3 builds on Topic 2:

- Functions in, and solutions to, the governing ODEs in Topics 1 and 2 are generally continuous – or at least, we focus on intervals where these functions are continuous.
- In Topic 3, we tackle a new class of ODEs where discontinuous and impulsive functions are involved.

Topic 3: Laplace transforms

Leaning outcomes:

1. Understand how ODEs involving discontinuous and impulsive functions can arise in modelling of real-world problems

2. Know how to calculate Laplace transforms and inverse Laplace transforms of appropriate functions

3. Know when and how to use the Laplace transform method for solving ODEs and IVPs

Lecture 10

We will:

1. Revise definition and calculation of improper integrals
2. Learn to use comparison test to determine convergence of an integral
3. Learn the definition of the Laplace transform and its basic properties
4. Discuss application of the Laplace transform to piecewise continuous functions

Improper integrals

$$\int_a^b f(t)\,\mathrm{d}t \qquad \text{(where we assume } a < b\text{)}$$

The above definite integral is *improper* if:

1. (a, b) is an infinite interval;
2. or f is not bounded on (a, b).

- Assume all unboundedness occurs at a and/or b (otherwise split up the integral).
- The improper integral is then defined as

$$\int_a^b f(t)\,\mathrm{d}t = \lim_{L\to a^+} \lim_{R\to b^-} \int_L^R f(t)\,\mathrm{d}t$$

Note that these limits are required to exist separately.

Exercise 3.1: Evaluate the following improper integrals

$$\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t \quad \text{and} \quad \int_{-1}^1 \frac{1}{t^2}\,\mathrm{d}t$$

For the first integral

$$\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t = \lim_{R\to\infty}\int_0^R \mathrm{e}^{-t}\,\mathrm{d}t = \lim_{R\to\infty}\left[-\mathrm{e}^{-t}\right]_0^R = -\lim_{R\to\infty}\{\mathrm{e}^{-R}-1\} = 1$$

For the second integral

$$\int_{-1}^1 \frac{1}{t^2}\,\mathrm{d}t = \int_{-1}^0 \frac{1}{t^2}\,\mathrm{d}t + \int_0^1 \frac{1}{t^2}\,\mathrm{d}t = \lim_{R\to 0^-}\int_{-1}^R \frac{1}{t^2}\,\mathrm{d}t + \lim_{L\to 0^+}\int_L^1 \frac{1}{t^2}\,\mathrm{d}t$$

$$= \lim_{R\to 0^-}\left[-\frac{1}{t}\right]_{-1}^R + \lim_{L\to 0^+}\left[-\frac{1}{t}\right]_L^1 \qquad \ldots \text{does not exist}$$

Comparison test

Definition (Comparison test)

Let f and g be two continuous functions on (a, b).

1. If $\big|f(t)\big| \leqslant g(t)$ for all $t \in (a, b)$ then $\displaystyle\int_a^b g(t)\,\mathrm{d}t$ converges $\Rightarrow$ $\displaystyle\int_a^b f(t)\,\mathrm{d}t$ converges

2. If $f(t) \geqslant g(t) \geqslant 0$ for all $t \in (a, b)$ then $\displaystyle\int_a^b g(t)\,\mathrm{d}t$ diverges $\Rightarrow$ $\displaystyle\int_a^b f(t)\,\mathrm{d}t$ diverges

Exercise 3.2: Does $\displaystyle\int_0^\infty \{\sin(t)\,\mathrm{e}^{-t}\}\,\mathrm{d}t$ converge?

$$\left.\begin{array}{l} \big|\sin(t)\,\mathrm{e}^{-t}\big| \leqslant \mathrm{e}^{-t} \quad \text{for} \quad t \in (0, \infty) \\ \displaystyle\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t \quad \text{converges (Ex 3.1a)} \end{array}\right\} \quad \int_0^\infty \{\sin(t)\,\mathrm{e}^{-t}\}\,\mathrm{d}t \quad \text{converges}$$

The Laplace transform

Definition (Laplace transform)

The *Laplace transform*, $\mathscr{L}$, of the function $f(t)$, where $t \in [0, \infty)$, is

$$\mathscr{L}\{f\} = \int_0^\infty \mathrm{e}^{-st} f(t)\, \mathrm{d}t = F(s)\ ,$$

defined for all $s \in \mathbb{C}$ for which this integral converges.

- Maps from t-space to s-space, where s is a complex-valued variable
- It is a linear transformation

Basic properties

Since it is a *linear transformation*:

1. $\mathscr{L}\{f + g\} = \mathscr{L}\{f\} + \mathscr{L}\{g\} = F(s) + G(s)$
2. $\mathscr{L}\{\beta f\} = \beta \mathscr{L}\{f\} = \beta F(s)$ where β is a constant

Exercise 3.3: Find the Laplace transform of $f(t) = 1$

$$\mathcal{L}\{1\} = \int_0^\infty \mathrm{e}^{-s\,t} \cdot 1 \,\mathrm{d}t = \lim_{R\to\infty} \int_0^R \mathrm{e}^{-s\,t}\,\mathrm{d}t$$

$$= \lim_{R\to\infty} \left[-\frac{1}{s}\,\mathrm{e}^{-s\,t} \right]_0^R = -\frac{1}{s} \lim_{R\to\infty} \left[\mathrm{e}^{-s\,R} - 1 \right] = \frac{1}{s}$$

Thus $\mathcal{L}\{1\} = F(s) = 1/s$.

The Laplace transform exists (integral converges) if we restrict to $\mathrm{Re}(s) > 0$. Indeed,

$$\left|\mathrm{e}^{-sR}\right| = \left|\mathrm{e}^{-R\,\mathrm{Re}(s) - iR\,\mathrm{Im}(s)}\right| = \mathrm{e}^{-R\,\mathrm{Re}(s)}$$

Exercise 3.4: Find the Laplace transform of $f(t) = \mathrm{e}^{\alpha t}$ for $\alpha \in \mathbb{R}$

$$\mathcal{L}\{\mathrm{e}^{\alpha t}\} = \int_0^\infty \mathrm{e}^{-s\,t}\,\mathrm{e}^{\alpha t}\,\mathrm{d}t = \lim_{R\to\infty} \int_0^R \mathrm{e}^{(\alpha-s)\,t}\mathrm{d}t$$

$$= \lim_{R\to\infty} \left[\frac{\mathrm{e}^{(\alpha-s)\,t}}{\alpha - s}\right]_0^R = \frac{1}{s-\alpha}$$

Thus $\mathcal{L}\{1\} = F(s) = 1/(s-\alpha)$ for $\mathrm{Re}(s) > \alpha$. The previous result is recovered for $\alpha = 0$.

The Laplace transform exists (integral converges) if we restrict to $\mathrm{Re}(s) > 0$. Indeed,

$$\left|\mathrm{e}^{(\alpha-s)R}\right| = \left|\mathrm{e}^{R\,\mathrm{Re}(\alpha-s)+iR\,\mathrm{Im}(\alpha-s)}\right| = \mathrm{e}^{-R\,[\mathrm{Re}(s)-\alpha]}$$

Existence of Laplace transform / Convergence of improper integral

Criteria to ensure the Laplace transform exists (improper integral converges)

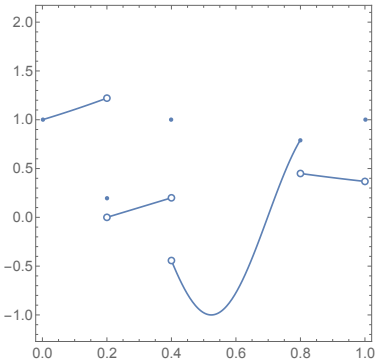
- f is piecewise continuous on $[0, L]$ for all $L > 0$
- There exist $\alpha, k, T \in \mathbb{R}$ such that $\left|f(t)\right| \leqslant k\,\mathrm{e}^{\alpha t}$ for all $t > T$
  - i.e., f bounded by an exponential function

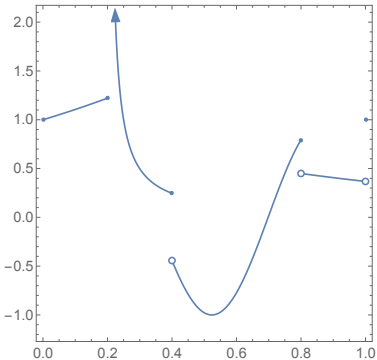
Then the Laplace transform exists and is guaranteed to converge for $\mathrm{Re}(s) > \alpha$

Piecewise continuous

- f has a finite number of discontinuities on $(0, L)$.
- The limits of f at the end points, and on each side of every discontinuity, are finite.

Exercise 3.5: Are the following functions piecewise continuous?





Exercise 3.6: Does the below function have a Laplace transform?

$$f = \begin{cases} 0\,, & 0 \leqslant t < 1 \\ 1\,, & 1 \leqslant t \end{cases}$$

- f contains one discontinuity (at $t = 1$)
- Limits at end points are bounded
- Therefore, it is piecewise continuous
- It is exponentially bounded for large t (with $\alpha = 0$)
- We deduce that f has a Laplace transform